

Travel Booking SQL-RAG Schema Knowledge Base

RAG-ready schema documentation for a natural-language-to-SQL application using MySQL.

Purpose: This document is intended to be chunked, embedded, retrieved, and supplied as schema context to an SQL-generating LLM. It explicitly documents tables, columns, relationships, join paths, business meaning, constraints, and common query patterns.

1. Database Overview

Database: travel_rag

Database type: MySQL

Primary use case: Travel booking analytics and natural-language SQL generation.

The schema contains five tables. **user_bookings** is the central transaction table. Users own bookings; bookings reference flights and optionally hotels; baggage records belong to bookings.

Table	Role	Primary Key	Important Foreign Keys
users	Customer master	user_id	—
user_bookings	Booking / transaction	booking_id	user_id, flight_id, hotel_id
flights	Flight master	flight_id	—
hotels	Hotel master	hotel_id	—
baggage_details	Booking baggage	baggage_id	booking_id

2. Relationship Graph

users 1 → N user_bookings; flights 1 → N user_bookings; hotels 1 → N user_bookings; user_bookings 1 → N baggage_details. hotel_id in user_bookings is nullable, so a booking may have no hotel.

Canonical join paths:
users.user_id = user_bookings.user_id
flights.flight_id = user_bookings.flight_id
hotels.hotel_id = user_bookings.hotel_id
user_bookings.booking_id = baggage_details.booking_id

3.1 Table: users

Business role: Customer master table.

Column	Type	Key	Nullable	Meaning
user_id	INT	PK	NO	Unique customer identifier.
first_name	VARCHAR(100)	—	NO	Customer first name.
last_name	VARCHAR(100)	—	NO	Customer last name.
email	VARCHAR(255)	UNIQUE	NO	Customer email address.
phone	VARCHAR(20)	—	YES	Customer phone number.
country	VARCHAR(100)	—	YES	Customer country.
created_at	TIMESTAMP	—	YES	Customer creation timestamp.

3.2 Table: user_bookings

Business role: Central booking transaction table.

Column	Type	Key	Nullable	Meaning
booking_id	INT	PK	NO	Unique booking identifier.

Column	Type	Key	Nullable	Meaning
user_id	INT	FK → users.user_id	NO	Customer who made the booking.
flight_id	INT	FK → flights.flight_id	NO	Flight associated with booking.
hotel_id	INT	FK → hotels.hotel_id	YES	Optional hotel associated with booking.
booking_date	DATETIME	—	YES	Timestamp when booking was created.
travel_date	DATE	—	NO	Travel date.
booking_status	VARCHAR(30)	—	NO	Booking state, e.g. CONFIRMED or CANCELLED.
number_of_passengers	INT	—	NO	Passengers represented by the booking.
total_amount	DECIMAL(10,2)	—	NO	Total booking amount.
currency	VARCHAR(3)	—	YES	Currency code, e.g. USD.

3.3 Table: flights

Business role: Flight master table.

Column	Type	Key	Nullable	Meaning
flight_id	INT	PK	NO	Unique flight identifier.
flight_number	VARCHAR(20)	UNIQUE	NO	Airline flight number.
airline	VARCHAR(100)	—	NO	Operating airline.
origin	VARCHAR(100)	—	NO	Departure location.
destination	VARCHAR(100)	—	NO	Arrival location.
departure_time	DATETIME	—	NO	Scheduled departure timestamp.
arrival_time	DATETIME	—	NO	Scheduled arrival timestamp.
seat_capacity	INT	—	NO	Total aircraft seat capacity.

3.4 Table: hotels

Business role: Hotel master table.

Column	Type	Key	Nullable	Meaning
hotel_id	INT	PK	NO	Unique hotel identifier.
hotel_name	VARCHAR(200)	—	NO	Hotel name.
city	VARCHAR(100)	—	NO	Hotel city.
country	VARCHAR(100)	—	NO	Hotel country.
star_rating	DECIMAL(2,1)	—	YES	Hotel rating.
price_per_night	DECIMAL(10,2)	—	NO	Nightly hotel price.

3.5 Table: baggage_details

Business role: Baggage details associated with a booking.

Column	Type	Key	Nullable	Meaning
baggage_id	INT	PK	NO	Unique baggage record identifier.
booking_id	INT	FK → user_bookings.booking_id	NO	Booking to which baggage belongs.
baggage_type	VARCHAR(30)	—	NO	Baggage category, e.g. CHECKED or CABIN.
weight_kg	DECIMAL(5,2)	—	NO	Weight represented by one baggage unit.

Column	Type	Key	Nullable	Meaning
baggage_count	INT	—	YES	Number of baggage units.
extra_charge	DECIMAL(10,2)	—	YES	Additional baggage charge.

4. Join Rules for SQL Generation

Relationship	SQL Join Condition	Cardinality	Use when
User → Booking	u.user_id = b.user_id	1:N	Need customer + booking information.
Booking → Flight	b.flight_id = f.flight_id	N:1	Need flight details for a booking.
Booking → Hotel	b.hotel_id = h.hotel_id	N:1 optional	Need hotel details. Prefer LEFT JOIN because hotel_id ca
Booking → Baggage	b.booking_id = bd.booking_id	1:N	Need baggage records or baggage aggregates.

5. Important SQL Generation Rules

- Use table aliases for multi-table queries: users u, user_bookings b, flights f, hotels h, baggage_details bd.
- Join users to bookings using u.user_id = b.user_id.
- Join bookings to flights using b.flight_id = f.flight_id.
- Join bookings to hotels using b.hotel_id = h.hotel_id. Because hotel_id is nullable, use LEFT JOIN when bookings without hotels must remain in the result.
- For baggage totals, aggregate at booking level. A common expression is SUM(bd.weight_kg * bd.baggage_count).
- When filtering booking state, use user_bookings.booking_status.
- When asking about travel timing, distinguish booking_date from travel_date and flight departure_time/arrival_time.
- Do not invent columns. If the requested concept is not represented in this schema, state that the schema does not contain the required field.
- Do not use SELECT * when a precise answer can be produced with explicit columns.
- SQL generation must be read-only for the RAG application. Prefer SELECT statements and reject INSERT, UPDATE, DELETE, DROP, ALTER, TRUNCATE, CREATE, and GRANT statements.

6. Business Vocabulary / Synonyms

User question term	Schema mapping
customer / passenger / traveler	users
customer ID / user ID	users.user_id
booking / reservation	user_bookings
booking status	user_bookings.booking_status
travel date	user_bookings.travel_date
booking date / reservation date	user_bookings.booking_date
flight	flights
airline	flights.airline
flight number	flights.flight_number
departure city / origin	flights.origin
arrival city / destination	flights.destination
hotel / accommodation	hotels
hotel rating / stars	hotels.star_rating
hotel price	hotels.price_per_night
luggage / baggage	baggage_details
checked baggage	baggage_details.baggage_type = 'CHECKED'
cabin baggage / hand luggage	baggage_details.baggage_type = 'CABIN'

User question term	Schema mapping
baggage weight	baggage_details.weight_kg
baggage fee / extra baggage charge	baggage_details.extra_charge